

Digestive system

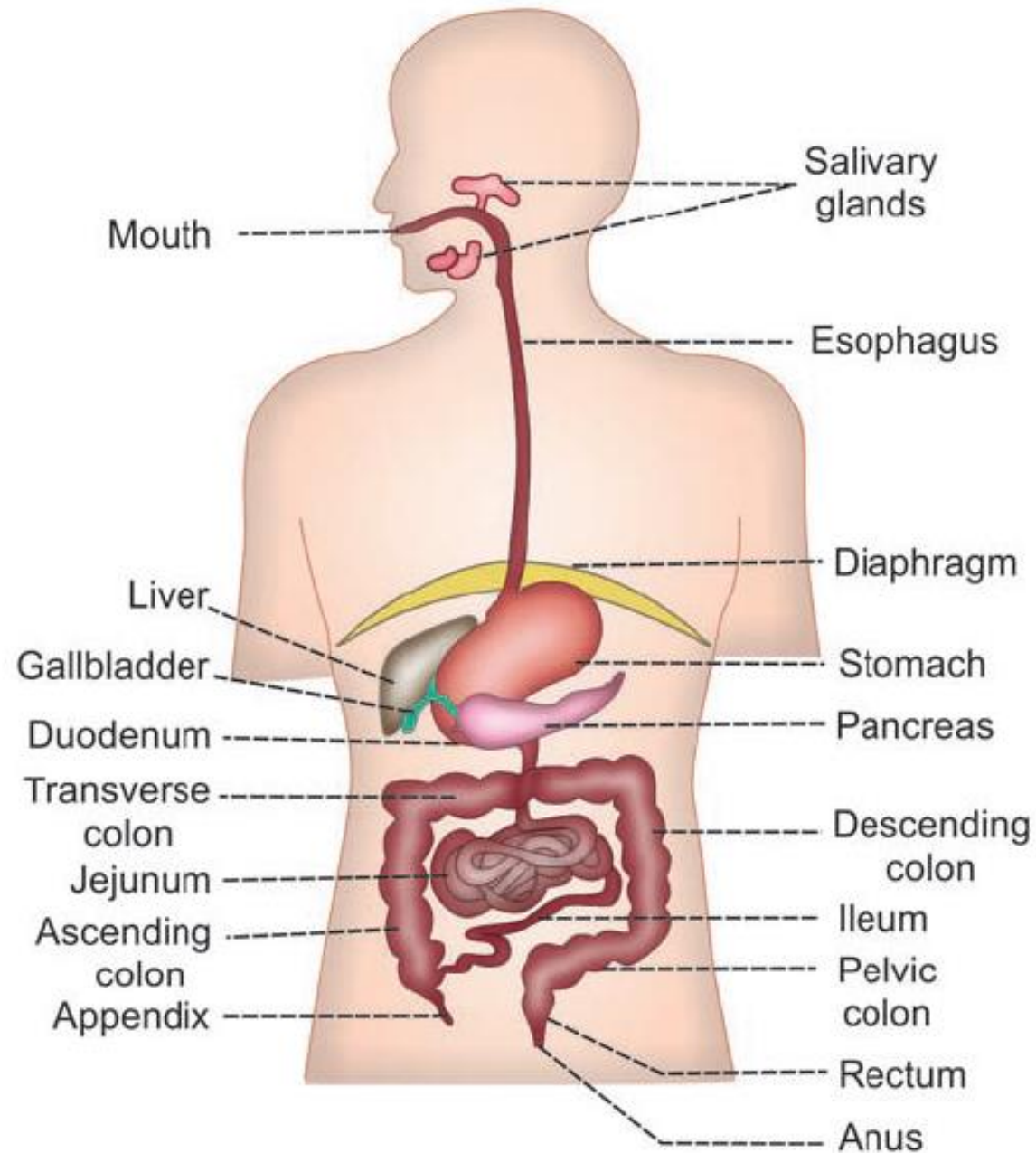
Dr Geethu Krishna C.G

Introduction to Digestive System

DIGESTION :

- Digestion is defined as the process by which food is broken down into simple chemical substances that can be absorbed and used as nutrients by the body

The GI Tract



Primary Digestive Organs

- i. Mouth
- ii. Pharynx
- iii. Esophagus
- iv. Stomach
- v. Small intestine
- vi. Large intestine

Accessory Digestive Organs

- i. Teeth
- ii. Tongue
- iii. Salivary glands
- iv. Exocrine part of pancreas
- v. Liver
- vi. Gallbladder.

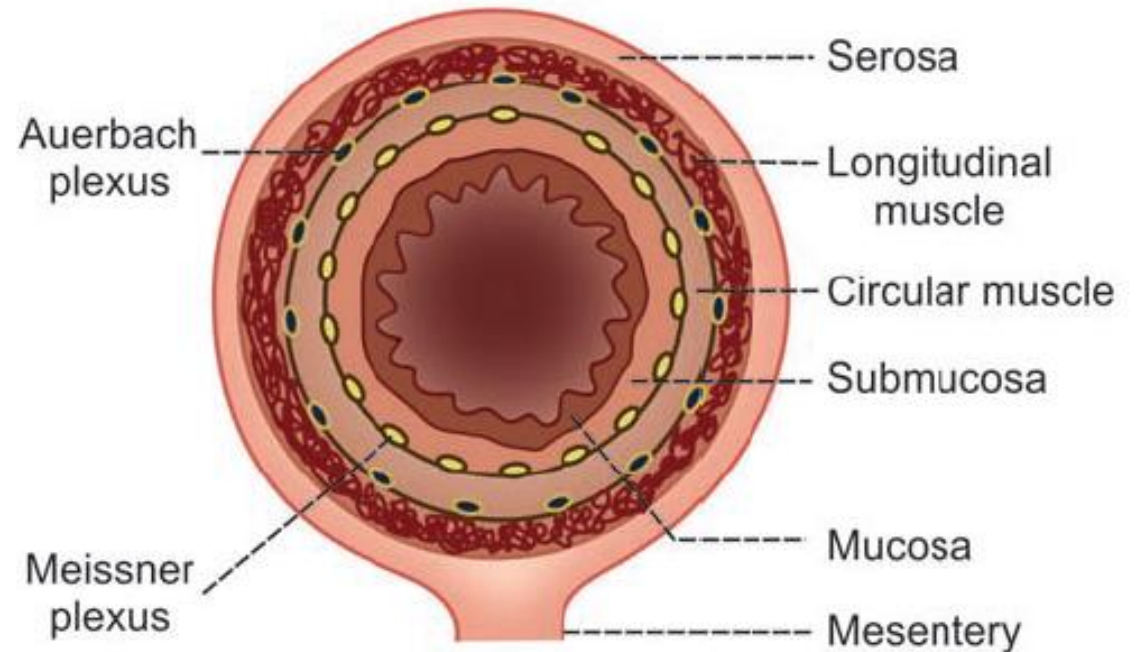
The functions of digestive system include:

1. Ingestion or consumption of food substances
2. Breaking them into small particles
3. Transport of small particles to different areas of the digestive tract
4. Secretion of necessary enzymes and other substances for digestion
5. Digestion of the food particles
6. Absorption of the digestive products (nutrients)
7. Removal of unwanted substances from the body.

WALL OF GASTROINTESTINAL TRACT

- In general, wall of the GI tract is formed by four layers which are from inside out:

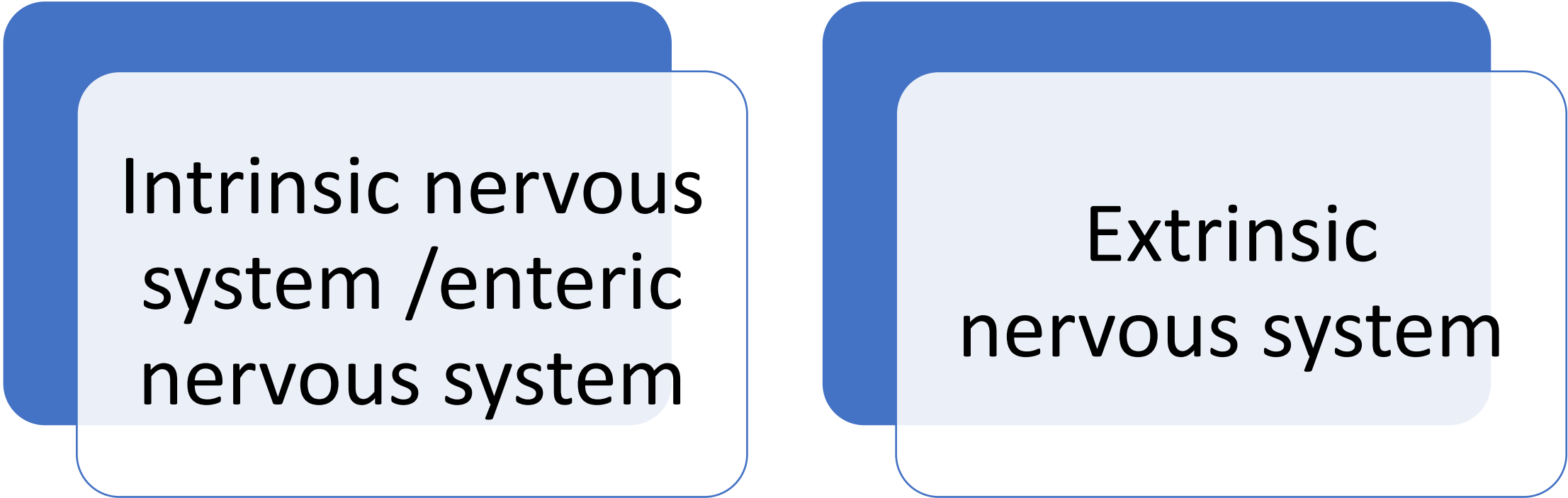
1. Mucus layer
2. Sub mucus layer
3. Muscular layer
4. Serous or fibrous layer.



- The epithelium which lines the lumen of the gut is thrown into series of small finger like projections called villi → site through which the end products of digestion are absorbed → they increase the surface area of the intestine → increases absorption

- 2 types of smooth muscles in the gut
- inner circular - Circular muscle encircles the gut ,important for function of sphincters in the gut
 - lower esophageal sphincter
 - pyloric sphincter
 - anal sphincter
- Outer longitudinal –responsible for peristalsis of gut

Innervation of gut



The diagram consists of two side-by-side boxes. Each box has a dark blue header and a light blue body. The left box contains the text 'Intrinsic nervous system /enteric nervous system' and the right box contains 'Extrinsic nervous system'. Both boxes are enclosed in a larger, light blue rounded rectangle.

Intrinsic nervous system /enteric nervous system

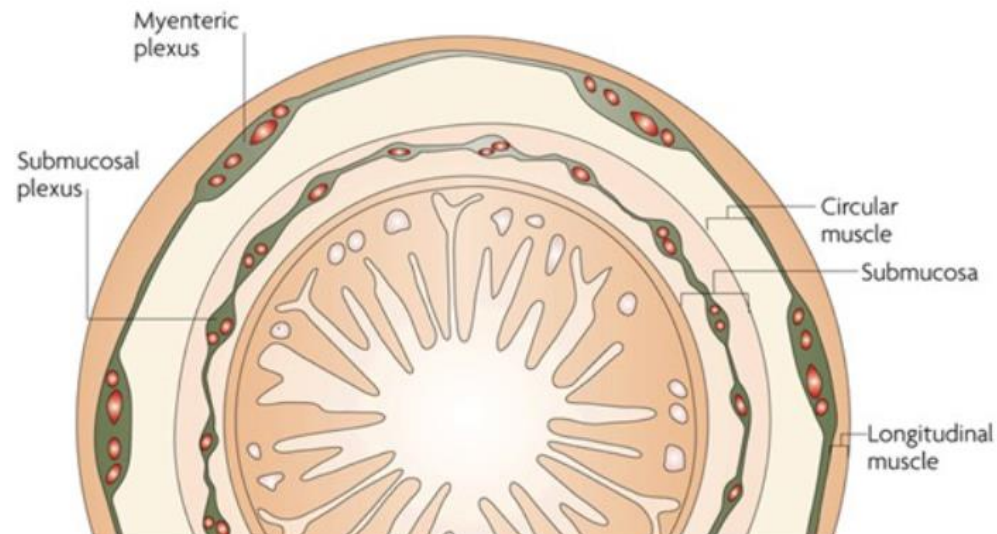
Extrinsic nervous system

Intrinsic nervous system/enteric nervous system

- Forms the local innervation of gut
 1. Auerbach's plexus of nerves /myenteric plexus
 2. Meissner's plexus

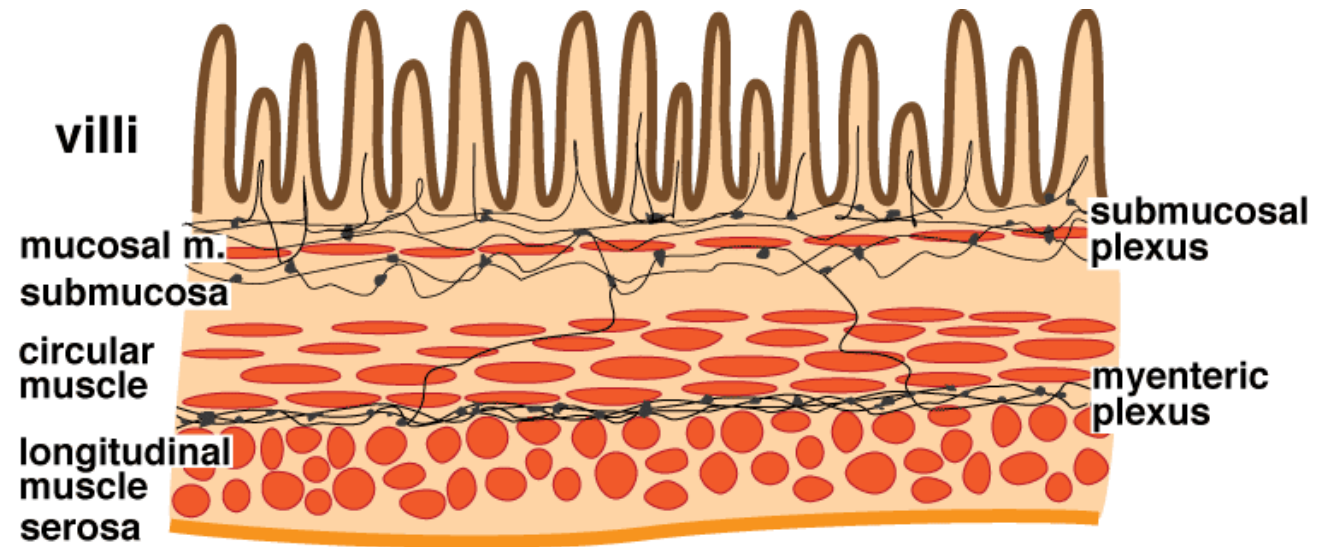
[Enteric (Intrinsic) nervous system]

- Submucosal plexus (Meissner's plexus)
- Myenteric plexus (Auerbach's plexus)



Auerbach's plexus/myenteric plexus

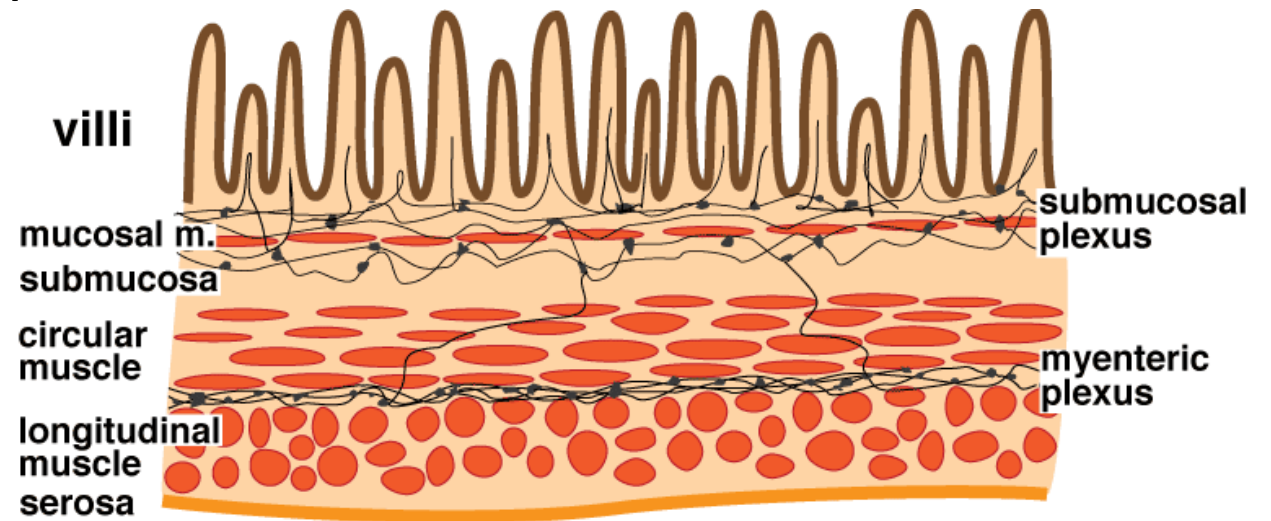
- Located in between longitudinal and muscular layers of the gut
- Regulate tone of the gut & motility



Transverse Section of Intestinal Wall

Meissner's plexus

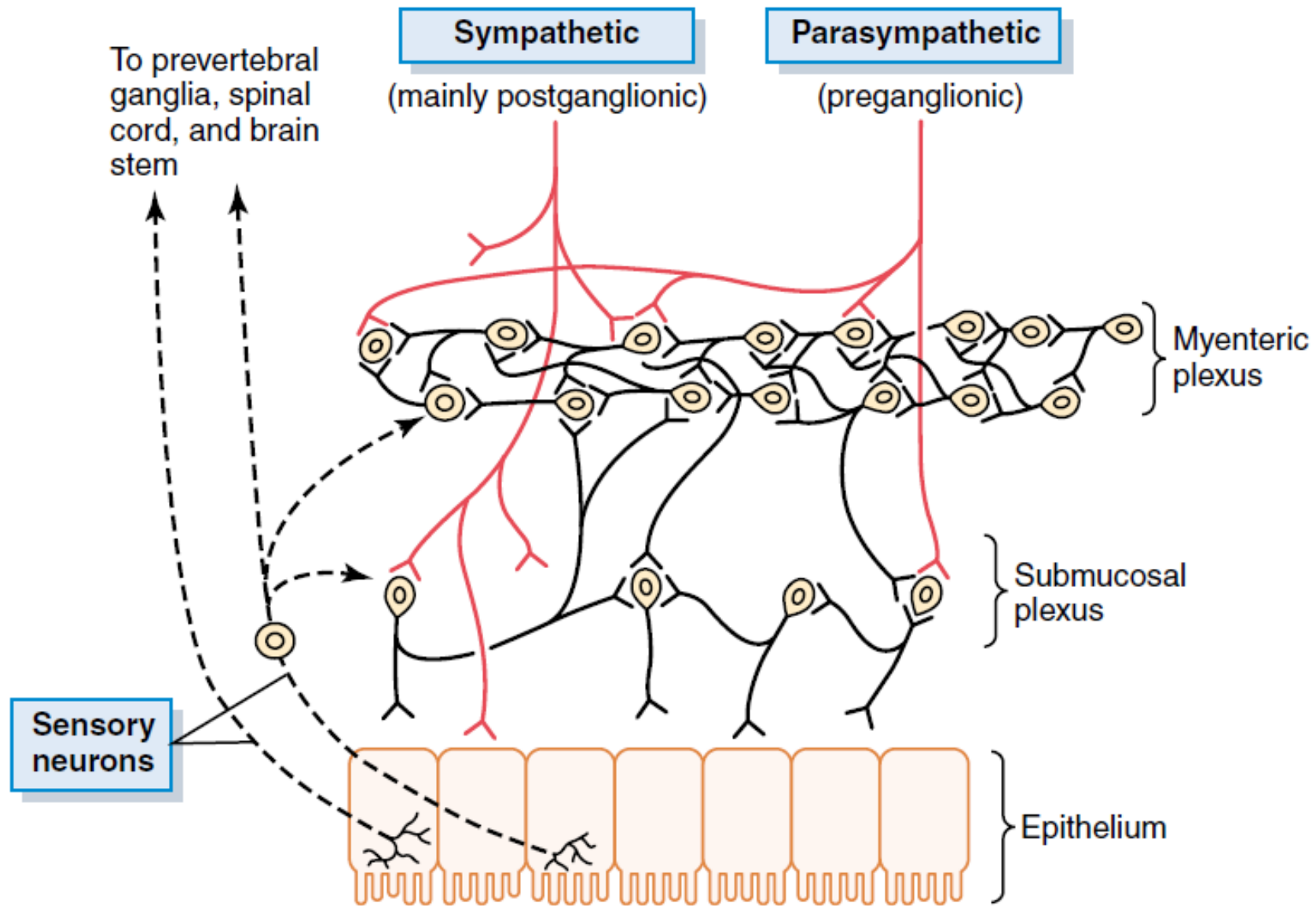
- Lies in the submucous layer of the gut
- Controls intestinal secretions and absorption



Transverse Section of Intestinal Wall

Extrinsic nervous system /autonomic nervous system

- Sympathetic ns-inhibit motility of gut ,increases contraction of sphincters
- Parasympathetic NS→ increases gut secretions and gut motility



Secretions of the gastrointestinal tract

SALIVARY GLANDS

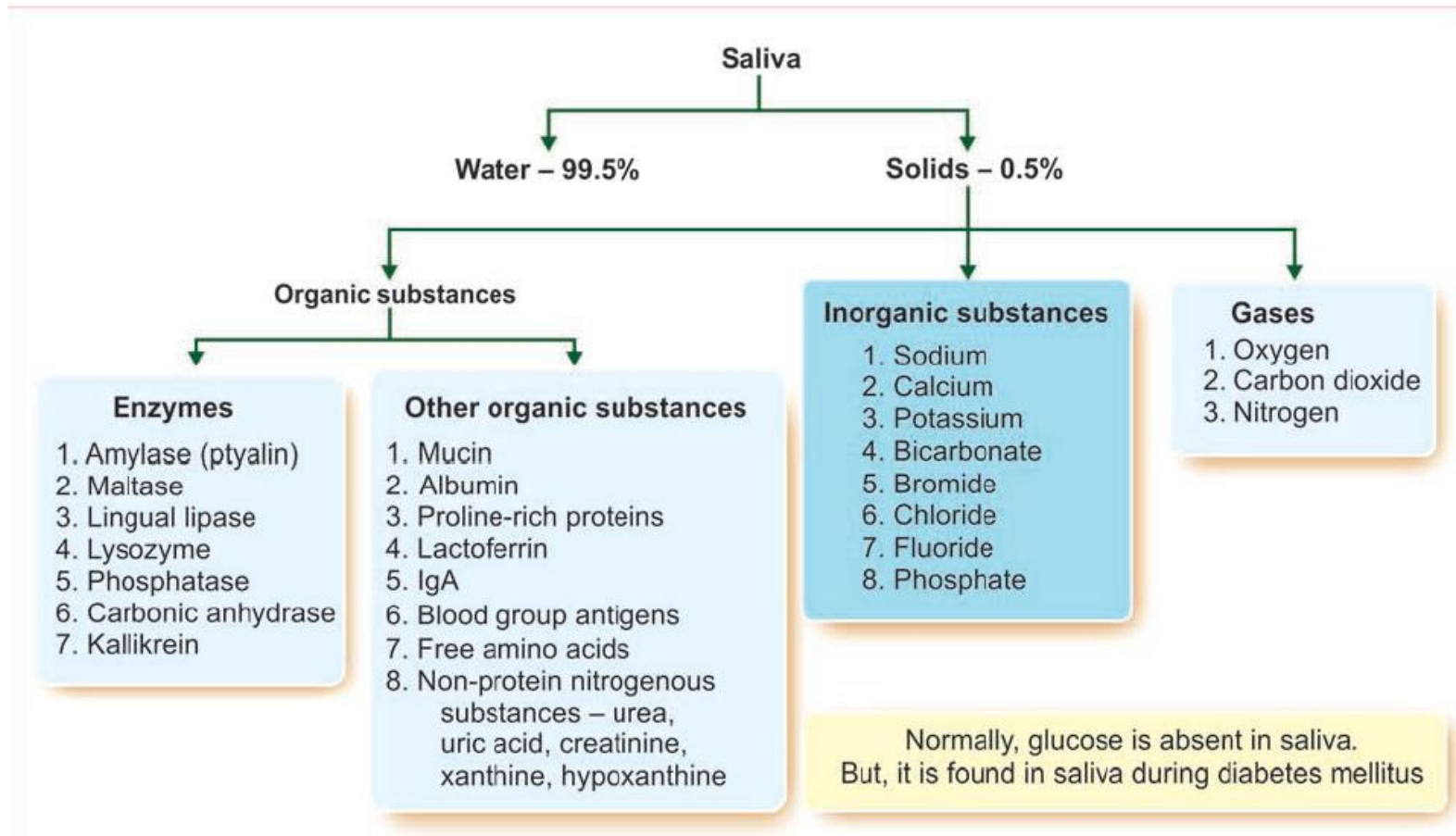
- In humans, the saliva is secreted by three pairs of major (larger) salivary glands and some minor (small) salivary glands.

MAJOR SALIVARY GLANDS

1. Parotid glands
2. Submaxillary or submandibular glands
3. Sublingual glands.

PROPERTIES OF SALIVA

- *Volume:* 1000 mL of saliva is secreted per day



Functions of saliva

1. It keeps the mouth moist and helps in lubrication → Formation of bolus ,speech
2. It helps in digestion through the action of salivary amylase
3. saliva is protective –it destroy viruses ,bacteria ,fungi
4. because of high bicarbonate content it acts as buffer

NERVE SUPPLY TO SALIVARY GLAND

- Salivary glands are supplied by both parasympathetic and sympathetic divisions of autonomic nervous system
- Stimulation of parasympathetic fibers of salivary glands causes secretion of saliva with large quantity of water
- Stimulation of sympathetic fibers causes secretion of saliva, which is thick and rich in organic constituents

REFLEX REGULATION OF SALIVARY SECRETION

- Salivary secretion is regulated by nervous mechanism through reflex action.
- Salivary reflexes are of two types:
 - 1. Unconditioned reflex.
 - 2. Conditioned reflex.

REFLEX REGULATION OF SALIVARY SECRETION

- **Unconditioned reflex** is the inborn reflex that is present since birth.
- This reflex induces salivary secretion when any substance is placed in the mouth.
- **Conditioned reflex** is the one that is acquired by experience and it needs previous experience
- The stimuli for this reflex are the sight, smell, hearing or thought of food

- *Xerostomia*

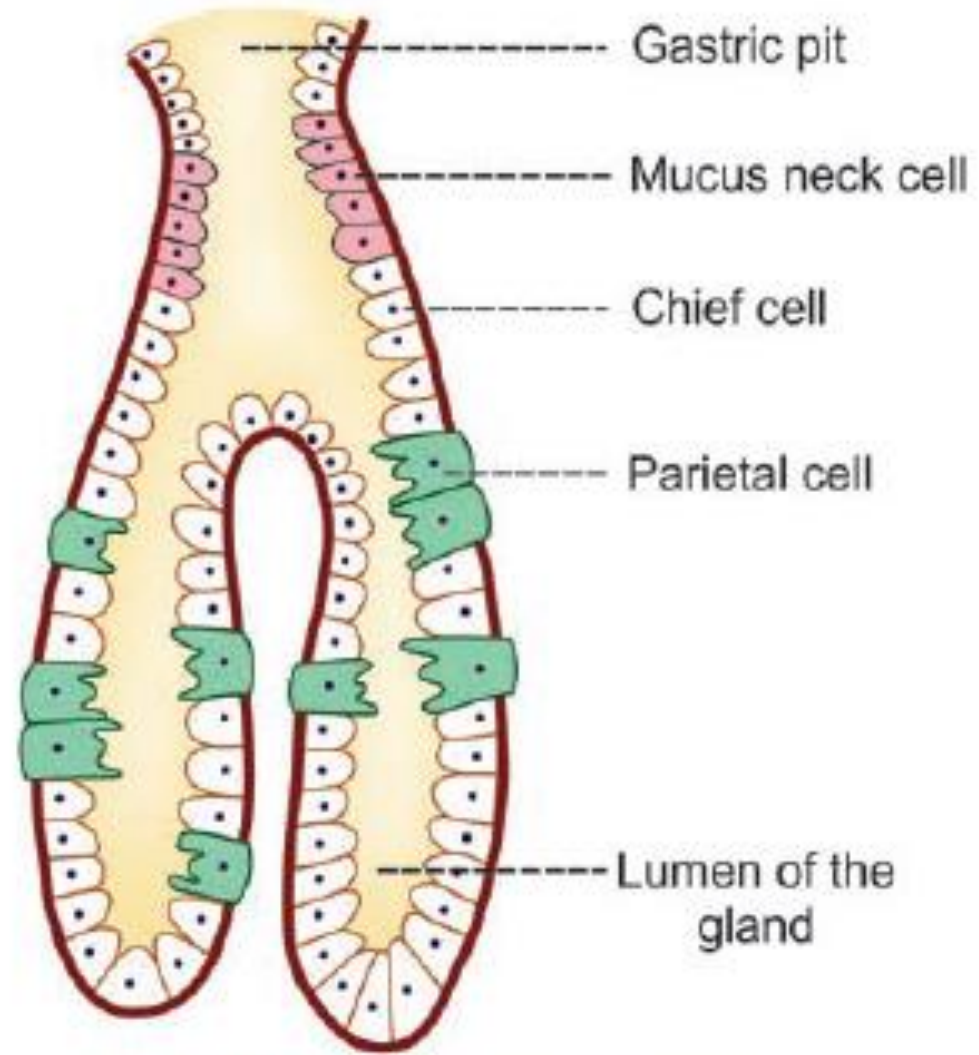
Xerostomia means dry mouth

Gastric juice secretion

Source

- Stomach secretes about 1.5-2L of gastric juice every day
- Secreted from gastric glands which mainly consists of 3 types of cells
 1. mucous neck cell
 2. peptic cell/chief cell
 3. parietal cell/OXYNTICcell
- 4. Enterochromaffin (EC) cells or Kulchitsky cells
- 5. 5. Enterochromaffin like (ECL) cells.

Gastric glands



Secretory function of cells in gastric gland

Cell	Secretory products
Chief cells	Pepsinogen Rennin Lipase Gelatinase Urase
Parietal cells	Hydrochloric acid Intrinsic factor of Castle
Mucus neck cells	Mucin
G cells	Gastrin
Enterochromaffin (EC) cells	Serotonin
Enterochromaffin-like (ECL) cells	Histamine

Components of gastric juice

- Water
- HCl-parietal cell
- Pepsin –chief cell
- Mucous –mucous neck cell
- IF –parietal cell

Functions of gastric juice

- large volumes of gastric juice allows the food to be mixed in stomach.
 - Food gets converted into a semifluid consistency which is acidic in nature → k/a acidic chyme
1. Digestive function
 2. Hemopoietic function
 3. Protective function

- HCL: the acid secreted by the parietal cells of the gastric glands
 1. it helps to kill bacteria and other microorganisms
 2. it converts pepsinogen to its active form ,pepsin
 3. it provide acidic Ph at which pepsin can act
- pepsinogen: proteolytic function –it digest protiens partly into peptides

- gastric lipase :it acts in an acidic environment
digestion of lipids

- Mucus:

Mucus, secreted by mucus neck cells of the gastric glands protects the stomach wall from irritation or mechanical injury

Protects the gastric mucosa from hydrochloric acid and pepsin digestion

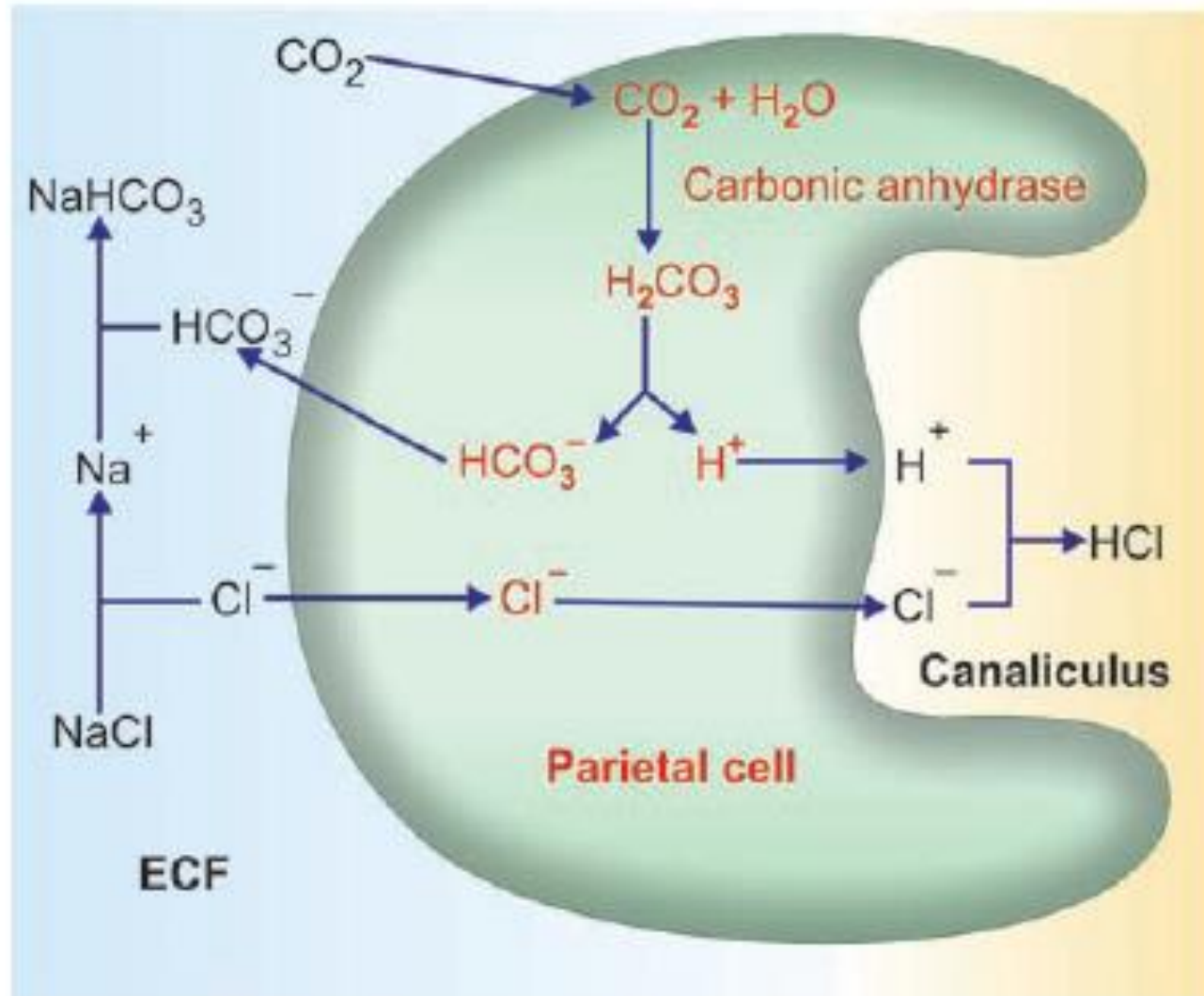
HEMOPOIETIC FUNCTION:-IF

- Intrinsic factor of Castle, secreted by parietal cells of gastric glands plays an important role in erythropoiesis.
- It is necessary for the absorption of vitamin B12 (which is called extrinsic factor) from GI tract into the blood. Vitamin B12 is an important maturation factor during erythropoiesis

SECRETION OF HYDROCHLORIC ACID

- According to **Davenport theory**, hydrochloric acid secretion is an **active process** that **takes place in the canaliculi of parietal cells** in gastric glands.
- The energy for this process is derived from oxidation of glucose.

SECRETION OF HYDROCHLORIC ACID



SECRETION OF HYDROCHLORIC ACID

- Carbon dioxide is derived from metabolic activities of parietal cell.
- Some amount of carbon dioxide is obtained from blood also.
- It combines with water to form **carbonic acid** in the presence of **carbonic anhydrase**.
- This enzyme is present in high concentration in parietal cells. Carbonic acid is the most unstable compound and immediately splits into hydrogen ion and bicarbonate ion.
- The hydrogen ion is actively pumped into the canaliculus of parietal cell.

SECRETION OF HYDROCHLORIC ACID

- Simultaneously, the chloride ion is also pumped into canaliculus actively.
- The chloride is derived from sodium chloride in the blood.
- Now, the hydrogen ion combines with chloride ion to form hydrochloric acid.
- To compensate the loss of chloride ion, the bicarbonate ion from parietal cell enters the blood and combines with sodium to form sodium bicarbonate.

SECRETION OF HYDROCHLORIC ACID

- *Factors Stimulating the Secretion of Hydrochloric Acid*

1. Gastrin
2. Histamine
3. Vagal stimulation.

SECRETION OF HYDROCHLORIC ACID

- *Factors Inhibiting the Secretion of Hydrochloric Acid*

1. Secretin
2. Gastric inhibitory polypeptide
3. Peptide YY.

Phases of gastric juice secretion

- Gastric secretion occurs in three different phases:

I. Cephalic phase

II. Gastric phase

III. Intestinal phase.

CEPHALIC PHASE :

- The thought ,sight,smell and taste of food can trigger the secretion of gastric juice
- This phase accounts for 30 % of gastric juice secretion
- Nervous mechanism regulates cephalic phase through reflex action.
- Two types of reflexes occur:
 1. Unconditioned reflex
 2. Conditioned reflex.

-

Conditioned Reflex-*Stages of reflex action:*

i. Impulses from the special sensory organs (eye, ear and nose) pass through afferent fibers of neural circuits to the cerebral cortex. Thinking of food stimulates the cerebral cortex directly

ii. From cerebral cortex, the impulses pass through dorsal nucleus of vagus and vagal efferents and reach the stomach wall

iii. Vagal nerve endings secrete acetylcholine, which stimulates the gastric secretion.

Sham feeding experiment to demonstrate initiation
of cephalic phase of gastric secretion



- Gastric phase :

- this phase secretion starts once the food enters the stomach
- It accounts for 60% of gastric juice secretion

- Intestinal phase :

this phase refers to continued secretion of gastric juice even after acid chyme has entered duodenum

Accounts for 10% gastric secretion

Nervous and hormonal control of gastric juice secretion

- Cephalic phase: under nervous control (vagus nerve)
- gastric phase : both nervous(Local myenteric reflex&vasovagal reflex)) and hormonal control(gastrin)
- intestinal phase : both nervous and hormonal control

Gastric mucosal barrier

- **DEFINITION** :The factors which tend to prevent injury to the gastric mucosa is known as gastric mucosal barrier
- It helps to maintain a difference in electrical potential and ionic concentration across the gastric mucosa.
- The gastric mucosal barrier protects the gastric mucosa from damage by intraluminal HCl, i.e. autodigestion.

Components of the mucosal barrier

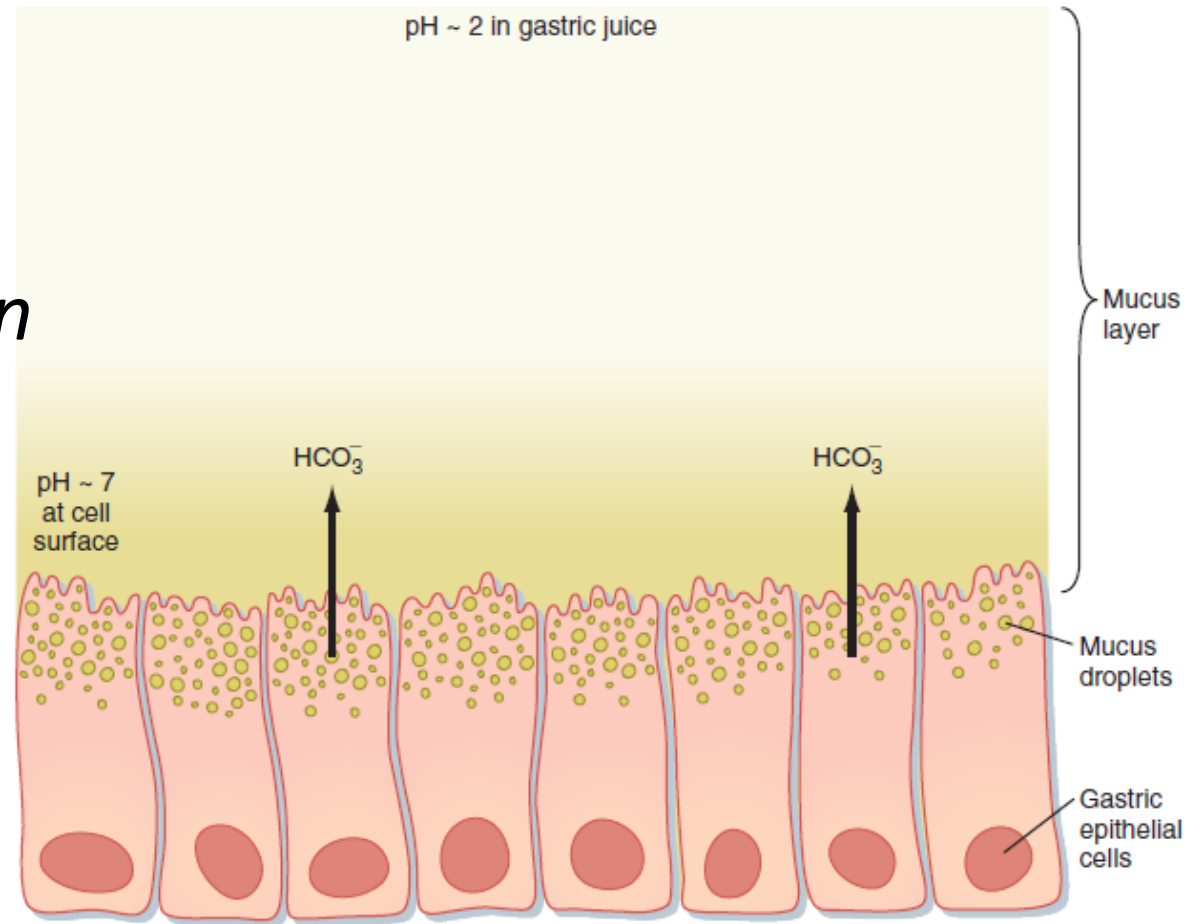
1. *Mucus layer*

Bicarbonate secretion, mucin

2. *Epithelial barrier.*

3. *High mucosal blood flow.*

4. *Prostaglandins*



PEPTIC ULCER

- Ulcer means the erosion of the surface of any organ
- If peptic ulcer is found in stomach, it is called **gastric ulcer** and if found in duodenum, it is called **duodenal ulcer**.
- **Causes**
 - i. Increased peptic activity due to excessive secretion of pepsin in gastric juice
 - ii. Hyperacidity of gastric juice
 - iii. Reduced alkalinity of duodenal content
 - iv. Decreased mucin content in gastric juice
 - v. Constant physical or emotional stress

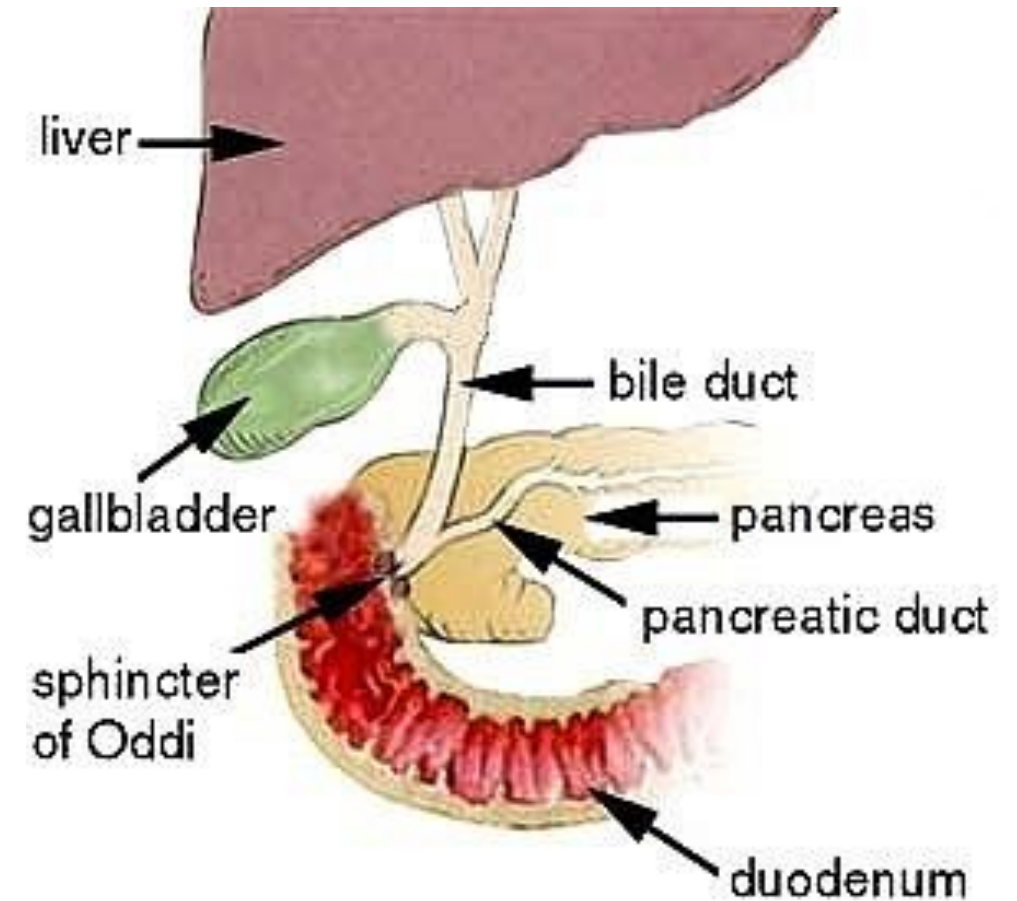
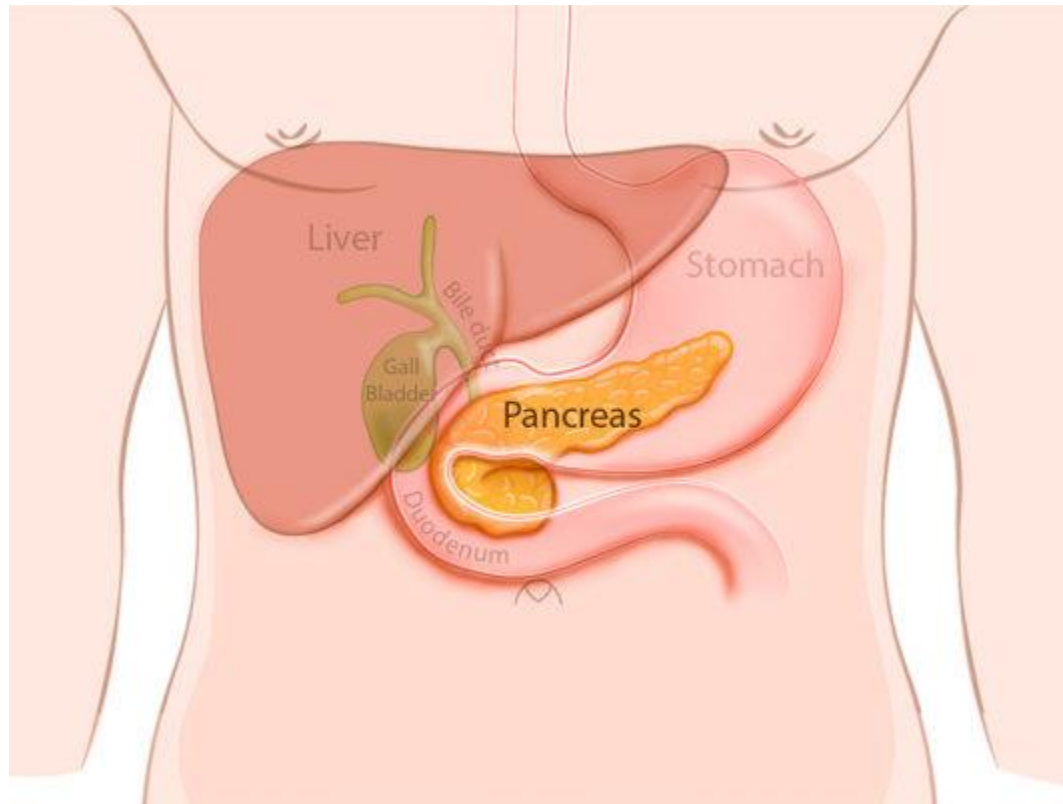
PEPTIC ULCER

- *Features*

- Most common feature of peptic ulcer is severe burning pain in epigastric region.
- i. Nausea
- ii. Vomiting
- iii. **Hematemesis** (vomiting blood)
- iv. **Heartburn** (burning pain in chest due to regurgitation of acid from stomach into esophagus)
- v. Anorexia (loss of appetite)
- vi. Loss of weight.

Pancreatic secretion

Pancreas



Pancreas

- Pancreas is a dual organ having two functions, namely **endocrine function** and **exocrine function**.
- Endocrine function is concerned with the production of hormones-insulin and glucagon
- The exocrine function is concerned with the secretion of digestive juice called pancreatic juice.

Composition of pancreatic juice

- Secretions from pancreas travel via pancreatic duct and open into second part of duodenum
- Pancreatic juice contains 99.5% of water and 0.5% of solids
- Contains enzymes for digestion of carbohydrates, proteins and lipids, HCO_3^-

Functions of HCO_3^- – in pancreatic juice

- High bicarbonate content makes the pancreatic juice **highly alkaline**, so that it protects the intestinal mucosa from acid chyme by neutralizing it.
- It makes the acid chyme into alkaline chyme
- Bicarbonate ions provide the required pH (7 to 9) for the activation of pancreatic enzymes